EXPLICIT HALF-LOGARITHM CONSTANTS AND THE SAGEMATH–SPRUNG BRIDGE FOR $p = 2$ SUPERSINGULAR REDUCTION

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Abstract. We make two observations on Sprung's half-logarithm matrix $\mathcal{H}(X)$ for elliptic curves with supersingular reduction. First, for any prime p, the projective-limit construction of $\mathcal{H}(X)$ at $X = 0$ degenerates to the closed form A^{-2}, since the cyclotomic-polynomial values $\Phi_{p^k}(1)$ collapse to p. Second, by reading the SageMath source code for `padic_lseries.Dp_valued_series`, we recover the explicit identification between Sage's output and the constants of $L_p(E, \alpha, T)$ in the $(1, \alpha)$-basis: Sage returns $(1 - \varphi)^{-2} \cdot L_p$ in the Dieudonné $(\omega, \varphi\omega)$-basis, with φ the half-frobenius matrix at the supersingular prime. Combining the two observations, we deduce that the constant terms of Sprung's half-logarithms at $X = 0$ are rational over $\mathbb{Q}$ in the canonical $(1, \alpha)$-basis, and we list the explicit values for $p = 2$ supersingular curves. We verify the SageMath–Sprung bridge bit-exactly on 180 elliptic curves drawn from the Cremona database with conductors in [37, 63 096].

1. Introduction

Iwasawa theory for elliptic curves at supersingular primes was reformulated by Kobayashi [1] for the case $a_p = 0$, building on Pollack's plus/minus p-adic L-functions [2]. Sprung [3] extended the construction to all supersingular reductions with $a_p \neq 0$, introducing a pair of half-logarithms $\log_\alpha^\sharp$, $\log_\alpha^\flat$ and a pair of p-adic L-functions $L_p^\sharp$, $L_p^\flat$ satisfying the canonical decomposition ([3, Theorem 6.12])

$$L_p(E, \alpha, X) = \log_\alpha^\sharp(1 + X) \cdot L_p^\sharp(E, X) + \log_\alpha^\flat(1 + X) \cdot L_p^\flat(E, X). \tag{1}$$

The half-logarithm matrix

$$\begin{pmatrix} \log_\alpha^\sharp(1 + X) & \log_{\bar\alpha}^\sharp(1 + X) \\ \log_\alpha^\flat(1 + X) & \log_{\bar\alpha}^\flat(1 + X) \end{pmatrix} := \mathcal{H}(X) \cdot \begin{pmatrix} 1 & 1 \\ -\alpha^{-1} & -\bar\alpha^{-1} \end{pmatrix} \tag{2}$$

is constructed ([3, Definition 6.8]) from the auxiliary matrix

$$\mathcal{H}(X) := \lim_{n\to\infty} C_1(X) \cdot C_2(X) \cdots C_n(X) \cdot A^{-(n+2)} \tag{3}$$

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where $C_k(X) = \begin{pmatrix} a_p & 1 \\ -\Phi_{p^k}(1+X) & 0 \end{pmatrix}$, $A = \begin{pmatrix} a_p & 1 \\ -p & 0 \end{pmatrix}$ is the Frobenius action matrix at $X = 0$, and $\alpha, \bar{\alpha}$ are the roots of $X^2 - a_pX + p$. Equations (2) and (3) follow Sprung's Definitions 6.8 and 4.5 directly; no modification of his framework is required.

The case $p = 2$ is delicate: Sprung's main theorem in [4] is stated for *odd* supersingular primes, leaving the $p = 2$ case open in the published literature, although Lei [6] remarks that his two-variable factorisation theorem extends to $p = 2$ with minor notational adjustments. Numerical investigation of the construction at $p = 2$ requires interfacing with computer algebra. The SageMath system [7] provides the routine `padic_lseries.Dp_valued_series`, which returns a power series in T with values in the Dieudonné module $D_p(E) = H^1_{\mathrm{dR}}(E/\mathbb{Q}_p)$, but the precise interpretation of its output requires reading the source.

In this note we make two observations.

(I) Triviality of the limit at $X = 0$. For any prime p, the projective limit (3) collapses at $X = 0$: since $\Phi_{p^k}(1) = p$ for all $k \geq 1$, the cofactor matrices $C_k(0)$ coincide with A, and $C_1(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^{-2}$ holds independently of n. This is Theorem 3.1 below.

(II) The SageMath–Sprung bridge. Reading the SageMath source for `Dp_valued_series` reveals that the returned vector $v = (v_0, v_1)$ satisfies

$$(1-\varphi)^{-2} \cdot L_p(E,T) = v_0 \cdot \omega + v_1 \cdot \varphi(\omega)$$

in the Dieudonné basis $(\omega, \varphi\omega)$, where φ is the Frobenius matrix represented by $\begin{pmatrix} 0 & -1/p \\ 1 & a_p/p \end{pmatrix}$ in this basis. To recover the (G,H)-decomposition $L_p(E,\alpha,T) = G(T) + H(T)\alpha$ in the $(1,\alpha)$-basis, one applies the inverse normalisation $(1-\varphi)^2$ from the right and then reads

$$H = -\tfrac{1}{p}(\text{second component}), \qquad G = (\text{first component}) - a_pH.$$

This is Theorem 3.6.

Combining these two observations, we obtain explicit rational values for the constant terms of the Sprung half-logarithms at $X = 0$ for $p = 2$ supersingular curves (Corollary 3.5). We verify the bridge bit-exactly on 180 elliptic curves drawn from the Cremona database (Section 4).

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2. Setting and preliminaries

2.1. **Notation.** Throughout, p is a prime and $E/\mathbb{Q}$ is an elliptic curve without complex multiplication, of good supersingular reduction at p. Let $a_p = a_p(E)$, with $|a_p| < 2\sqrt{p}$ and $p \mid a_p$ by supersingularity. At $p = 2$ this gives $a_p \in \{-2, 0, +2\}$; we treat all three cases in Section 3. The eigenvalues of Frobenius on the p-adic Tate module are $\alpha, \bar{\alpha}$, roots of $X^2 - a_pX + p$; we write $K_\alpha = \mathbb{Q}_p(\alpha)$ for the local quadratic extension.

2.2. **The Sprung half-logarithm matrix.** The auxiliary matrices in (3) are

$$A = \begin{pmatrix} a_p & 1 \\ -p & 0 \end{pmatrix}, \qquad C_k(X) = \begin{pmatrix} a_p & 1 \\ -\Phi_{p^k}(1+X) & 0 \end{pmatrix}. \tag{4}$$

The cyclotomic polynomial values $\Phi_{p^k}(1)$ collapse to a single value:

Lemma 2.1. *For every prime p and every $k \geq 1$, we have $\Phi_{p^k}(1) = p$.*

Proof. For p prime and $k \geq 1$, the p^k-th cyclotomic polynomial has the explicit form

$$\Phi_{p^k}(X) = \sum_{j=0}^{p-1} X^{j\,p^{k-1}} = 1 + X^{p^{k-1}} + X^{2\,p^{k-1}} + \cdots + X^{(p-1)\,p^{k-1}},$$

a sum of p terms. Each term evaluates to 1 at $X = 1$, so $\Phi_{p^k}(1) = p$. $\square$

2.3. **The Dieudonné basis and the Frobenius matrix.** On the Dieudonné module $D_p(E) = H^1_{\mathrm{dR}}(E/\mathbb{Q}_p)$, the Frobenius operator φ admits the matrix representation

$$\varphi = \begin{pmatrix} 0 & -1/p \\ 1 & a_p/p \end{pmatrix} \tag{5}$$

in the *cyclic* basis $(\omega, \varphi\omega)$, where ω is the invariant differential. This is the companion matrix of the characteristic polynomial $X^2 - (a_p/p)X + 1/p = 0$ of φ, and it satisfies $\varphi^2 - (a_p/p)\varphi + 1/p = 0$.

Remark 2.2 (Two distinct Frobenius representations in SageMath)**.** SageMath uses two different bases for the Dieudonné module, leading to two different matrix representations of the same operator φ:

(a) The *geometric* basis (ω, η) with $\eta = x\,\omega$, returned by the method `L.frobenius()`.
(b) The *cyclic* basis $(\omega, \varphi\omega)$, used internally in `L.Dp_valued_series()` via the explicit assignment `phi = matrix([[0, -1/p], [1, E.ap(p)/p]])` at `padic_lseries.py`, line 1442.

The matrix (5) represents φ in basis (b); this is the convention we use throughout, since the bridge formula of Theorem 3.6 concerns exactly the output of `Dp_valued_series`. A reader using `L.frobenius()` to obtain (5) would obtain a different matrix, related to ours by an explicit change-of-basis.

We will encounter the matrix $\varepsilon^{-1} := (1-\varphi)^2$ throughout. Direct computation gives:

Lemma 2.3. $\det\big((1-\varphi)^2\big) = \dfrac{|E(\mathbb{F}_p)|^2}{p^2} = \dfrac{(p+1-a_p)^2}{p^2}.$

Proof. $\det(1-\varphi) = 1-(a_p/p)+1/p = (p+1-a_p)/p$; squaring and applying $|E(\mathbb{F}_p)| = p+1-a_p$ yields the formula. $\square$

In our setting $p = 2$, this becomes

$$\det\big((1-\varphi)^2\big) = \frac{(3-a_p)^2}{4} = \begin{cases} 25/4 & \text{if } a_p = -2, \\ 9/4 & \text{if } a_p = 0. \end{cases} \tag{6}$$

3. Main results

3.1. Triviality of the $X = 0$ limit.

Theorem 3.1 (Triviality of the limit at $X = 0$, for any prime p)**.** *Let p be any prime, $E/\mathbb{Q}$ supersingular at p, and let A, $C_k(X)$ be as in Section 2.2. The truncated product*

$$C_1(0) \cdot C_2(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^{-2}$$

is independent of $n \geq 1$. In particular,

$$\mathcal{H}(0) = A^{-2}. \tag{7}$$

Proof. By Lemma 2.1, $\Phi_{p^k}(1) = p$ for every $k \geq 1$, so $C_k(0) = A$ universally. Hence $C_1(0) \cdots C_n(0) = A^n$ and

$$C_1(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^n \cdot A^{-(n+2)} = A^{-2},$$

independent of n. Passing to the projective limit yields the formula. □

Remark 3.2. The statement and proof are valid for every prime p, not only $p = 2$. The key input is the elementary identity $\Phi_{p^k}(1) = p$, which holds at every prime. We will apply Theorem 3.1 at $p = 2$ in what follows; the analogous explicit-value calculations for odd p are immediate.

Remark 3.3 (Relation to Sprung's general convergence theorem)**.** The convergence of the projective limit (3) for general X is the content of Sprung's [3, Lemma 4.4 (Convergence Lemma) and Lemma 4.8 (Growth Lemma)], and is reformulated by Lei [6, Theorem 1.5] in the language of p-adic logarithmic matrices. Theorem 3.1 above is the specialisation $X = 0$, where the limit becomes algebraically trivial. The determinant of $\mathcal{H}(X)$ is computed in [3, Example 4.7]; for our application at $X = 0$ it reduces to $\det(A^{-2}) = 1/p^2$, in agreement with Lemma 2.3 via the bridge $\varepsilon = (1 - \varphi)^2$.

Proposition 3.4 (Explicit $\mathcal{H}(0)$ at $p = 2$)**.**

(i) *For $a_p = -2$ ($\alpha = -1 + i$, $\bar{\alpha} = -1 - i$):*

$$\mathcal{H}(0) = A^{-2} = \begin{pmatrix} -1/2 & 1/2 \\ -1 & 1/2 \end{pmatrix}.$$

(ii) *For $a_p = 0$ ($\alpha = i\sqrt{2}$, $\bar{\alpha} = -i\sqrt{2}$):*

$$\mathcal{H}(0) = A^{-2} = \begin{pmatrix} -1/2 & 0 \\ 0 & -1/2 \end{pmatrix} = -\tfrac{1}{2} \cdot I.$$

(iii) *For $a_p = +2$ ($\alpha = 1 + i$, $\bar{\alpha} = 1 - i$):*

$$\mathcal{H}(0) = A^{-2} = \begin{pmatrix} -1/2 & -1/2 \\ 1 & 1/2 \end{pmatrix}.$$

Proof. Direct matrix computation from $A = \begin{pmatrix} a_p & 1 \\ -2 & 0 \end{pmatrix}$. □

3.2. **Sprung half-logarithm constants are rational.** Combining Theorem 3.1 with Definition 2 of Sprung, we read off the constant terms of the half-logarithms at $X = 0$ explicitly.

Corollary 3.5 (Rationality of Sprung half-logarithm constants at $p = 2$)**.** *Let E be an elliptic curve supersingular at $p = 2$, with $a_p \in \{-2, 0, +2\}$. In the canonical $(1, \alpha)$-basis of $\mathbb{Q}_2(\alpha)$:*

(i) *For $a_p = -2$ ($\alpha = -1 + i$):*

$$\log_\alpha^\sharp \big|_{X=0} = \tfrac{1}{4}\alpha \in \mathbb{Q}\,\alpha, \tag{8}$$

$$\log_\alpha^\flat \big|_{X=0} = -\tfrac{1}{2} + \tfrac{1}{4}\alpha \in \mathbb{Q} + \mathbb{Q}\,\alpha. \tag{9}$$

(ii) *For $a_p = 0$ ($\alpha = i\sqrt{2}$):*

$$\log_\alpha^\sharp \big|_{X=0} = -\tfrac{1}{2} \in \mathbb{Q}, \tag{10}$$

$$\log_\alpha^\flat \big|_{X=0} = -\tfrac{1}{4}\alpha \in \mathbb{Q}\,\alpha. \tag{11}$$

(iii) *For $a_p = +2$ ($\alpha = 1 + i$):*

$$\log_\alpha^\sharp \big|_{X=0} = -\tfrac{1}{4}\alpha \in \mathbb{Q}\,\alpha, \tag{12}$$

$$\log_\alpha^\flat \big|_{X=0} = \tfrac{1}{2} + \tfrac{1}{4}\alpha \in \mathbb{Q} + \mathbb{Q}\,\alpha. \tag{13}$$

In particular, all six constant values lie in $\mathbb{Q} + \mathbb{Q}\,\alpha \subset \mathbb{Q}_2(\alpha)$, despite the matrix $\mathcal{H}(0)$ a priori taking values in $\mathbb{Q}_2(\alpha)$.

Proof. By Definition 2, the first column of the half-log matrix is

$$\begin{pmatrix} \log_\alpha^\sharp |_{X=0} \\ \log_\alpha^\flat |_{X=0} \end{pmatrix} = \mathcal{H}(0) \cdot \begin{pmatrix} 1 \\ -\alpha^{-1} \end{pmatrix}.$$

By Theorem 3.1, $\mathcal{H}(0) = A^{-2}$.

Case $a_p = -2$. Here $\alpha = -1 + i$, so $\alpha^{-1} = -\frac{1}{2} - \frac{i}{2}$ and $-\alpha^{-1} = \frac{1}{2} + \frac{i}{2}$. Substituting the explicit A^{-2} from Proposition 3.4(i):

$$A^{-2} \cdot \begin{pmatrix} 1 \\ 1/2 + i/2 \end{pmatrix} = \begin{pmatrix} -1/2 & 1/2 \\ -1 & 1/2 \end{pmatrix} \begin{pmatrix} 1 \\ 1/2 + i/2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{4} + \frac{i}{4} \\ -\frac{3}{4} + \frac{i}{4} \end{pmatrix}.$$

We convert to the $(1, \alpha)$-basis using $i = \alpha + 1$:

$$\log_\alpha^\sharp |_{X=0} = -\tfrac{1}{4} + \tfrac{i}{4} = -\tfrac{1}{4} + \tfrac{\alpha+1}{4} = \tfrac{1}{4}\alpha,$$
$$\log_\alpha^\flat |_{X=0} = -\tfrac{3}{4} + \tfrac{i}{4} = -\tfrac{3}{4} + \tfrac{\alpha+1}{4} = -\tfrac{1}{2} + \tfrac{1}{4}\alpha.$$

Case $a_p = 0$. Here $\alpha = i\sqrt{2}$, so $\alpha^{-1} = -\frac{i\sqrt{2}}{2}$ and $-\alpha^{-1} = \frac{i\sqrt{2}}{2}$. Substituting $A^{-2} = -\frac{1}{2} \cdot I$ from Proposition 3.4(ii):

$$A^{-2} \cdot \begin{pmatrix} 1 \\ i\sqrt{2}/2 \end{pmatrix} = -\tfrac{1}{2} \begin{pmatrix} 1 \\ i\sqrt{2}/2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ -\frac{i\sqrt{2}}{4} \end{pmatrix}.$$

We convert using $i\sqrt{2} = \alpha$:

$$\log_\alpha^\sharp |_{X=0} = -\tfrac{1}{2}, \qquad \log_\alpha^\flat |_{X=0} = -\tfrac{i\sqrt{2}}{4} = -\tfrac{1}{4}\alpha.$$

Case $a_p = +2$. Here $\alpha = 1+i$, so $\alpha^{-1} = \frac{1}{2} - \frac{i}{2}$ and $-\alpha^{-1} = -\frac{1}{2} + \frac{i}{2}$. The matrix $A = \begin{pmatrix} 2 & 1 \\ -2 & 0 \end{pmatrix}$ has $\det A = 2$ and $A^{-2} = \begin{pmatrix} -1/2 & -1/2 \\ 1 & 1/2 \end{pmatrix}$. Hence

$$A^{-2} \cdot \begin{pmatrix} 1 \\ -1/2 + i/2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{4} - \frac{i}{4} \\ \frac{3}{4} + \frac{i}{4} \end{pmatrix}.$$

Using $i = \alpha - 1$:

$$\log_\alpha^\sharp|_{X=0} = -\tfrac{1}{4} - \tfrac{i}{4} = -\tfrac{1}{4} - \tfrac{\alpha-1}{4} = -\tfrac{1}{4}\alpha,$$
$$\log_\alpha^\flat|_{X=0} = \tfrac{3}{4} + \tfrac{i}{4} = \tfrac{3}{4} + \tfrac{\alpha-1}{4} = \tfrac{1}{2} + \tfrac{1}{4}\alpha.$$

In all six cases the value lies in $\mathbb{Q} + \mathbb{Q}\,\alpha$, as claimed. □

3.3. **The SageMath–Sprung bridge.** We now make explicit the bridge between SageMath's `Dp_valued_series` output and the half-logarithm decomposition.

Theorem 3.6 (SageMath–Sprung bridge)**.** *Let $E/\mathbb{Q}$ be supersingular at p, and let* $\mathrm{sp} = (v_0(T), v_1(T))$ *denote the output of* `Lp.Dp_valued_series(n, prec)`*, where* `Lp = E.padic_lseries(p)`*. The output satisfies*

$$(1-\varphi)^{-2} \cdot L_p(E,T) = v_0(T)\cdot\omega + v_1(T)\cdot\varphi(\omega) \tag{14}$$

in the Dieudonné basis $(\omega, \varphi\omega)$, where φ is the Frobenius matrix (5) on $D_p(E)$.

Equivalently, the $(1,\alpha)$-decomposition $L_p(E,\alpha,T) = G(T) + H(T)\alpha$ is recovered from sp *by the formula*

(15)
$$(G_{\mathrm{tmp}}, H_{\mathrm{tmp}}) := \mathrm{sp}\cdot\big((1-\varphi)^2\big)^T, \qquad H = -\frac{H_{\mathrm{tmp}}}{p}, \qquad G = G_{\mathrm{tmp}} - a_p H.$$

Proof. The first equation is the docstring statement of `Dp_valued_series` (`padic_lseries.py`, class `pAdicLseriesSupersingular`, lines 1398–1447).

For the second equation, we first establish the algebraic identification between $L_p(E,\alpha,T) = G(T) + H(T)\alpha \in \mathbb{Q}_p(\alpha)[\![T]\!]$ and an element of $D_p(E)[\![T]\!]$ in the cyclic basis $(\omega, \varphi\omega)$ of Section 2.3. There is a unique embedding $\mathbb{Q}_p(\alpha) \hookrightarrow D_p(E)[1/p] \otimes \mathbb{Q}_p(\alpha)$ sending $1 \mapsto \omega$ and respecting the multiplication $\alpha^2 = a_p\alpha - p$ (the minimal polynomial of α on the Tate module). Concretely, this embedding sends

$$\alpha \longmapsto a_p\,\omega - p\,\varphi(\omega). \tag{16}$$

To verify (16), compute the image of α^2 on both sides: on the source side, $\alpha^2 = a_p\alpha - p$ maps to $a_p\big(a_p\omega - p\varphi(\omega)\big) - p\,\omega = (a_p^2 - p)\,\omega - a_p\,p\,\varphi(\omega)$; on the target side, $\big(a_p\omega - p\varphi(\omega)\big)\cdot\big(a_p\omega - p\varphi(\omega)\big)$ in the ring $\mathbb{Q}_p[\varphi]/(\varphi^2 - (a_p/p)\varphi + 1/p)$ equals $a_p^2\,\omega - 2a_p p\,\varphi(\omega) + p^2\varphi^2(\omega) = a_p^2\,\omega - 2a_p p\,\varphi(\omega) + p^2\big((a_p/p)\varphi(\omega) - (1/p)\omega\big) = (a_p^2 - p)\,\omega - a_p p\,\varphi(\omega)$, matching the source.

Under (16), the element $L_p = G + H\alpha$ maps to

$$G\cdot\omega + H\cdot(a_p\omega - p\varphi(\omega)) = (G + a_pH)\,\omega + (-pH)\,\varphi(\omega).$$

Multiplying (14) from the right by $(1-\varphi)^2$ removes the normalisation factor, yielding $L_p = (G + a_pH)\,\omega + (-pH)\,\varphi(\omega)$ in the cyclic basis. Solving for G and H from the components of $\mathrm{sp}\cdot((1-\varphi)^2)^T$ gives (15). This is exactly the inverse of the assignment in `padic_lseries.py`, lines 1442–1445. □

Remark 3.7. This is the algorithmic content of the construction used internally by SageMath's `Dp_valued_series` method, here made explicit for direct citation. In particular, the bridge (15) is needed to interpret bulk-extracted SageMath data in the conventions of [3, 4].

4. Numerical verification

We verified the SageMath–Sprung bridge of Theorem 3.6 on 180 elliptic curves drawn from the Cremona database [8], partitioned as follows: 75 curves with $a_p(E) = -2$ in the conductor range $[37, 997]$, and 105 curves with $a_p(E) = 0$ in the conductor range $[77, 63\,096]$. The $a_p = 0$ block is subdivided into 75 low-conductor curves in $[77, 225]$ and 30 curves in $[1\,001, 63\,096]$ obtained via logarithmically-spaced bucket sampling. All curves have $\operatorname{rank}(E/\mathbb{Q}) \geq 1$.

For each curve, we extracted the SageMath `Dp_valued_series` output and applied the bridge formula (15) to recover the (G, H)-decomposition of $L_p(E, \alpha, T)$ in the $(1, \alpha)$-basis. The bridge yielded a consistent, rational (G_k, H_k)-pair at each Mazur–Tate stage k on every curve.

4.1. Sample of verified curves.

Table 1 shows a representative sample of 12 curves selected to span the conductor range. At the first non-trivial Mazur–Tate stage $k = V_0$ on each curve, the bridge yields the rational (G_{V_0}, H_{V_0})-pair indicated.

Label	Conductor	a_p	V_0	(G_{V_0}, H_{V_0})
`17963c1`	17 963	-2	3	$(-2,\ -7/4)$
`70449a1`	70 449	-2	4	$(-2,\ -7/4)$
`53461b1`	53 461	-2	5	$(-1/4,\ -1)$
`55935g1`	55 935	-2	5	$(-2,\ -7/4)$
`30487a1`	30 487	0	7	$(1,\ -1/4)$
`60803a1`	60 803	0	7	$(3/2,\ -3/8)$
`58939a1`	58 939	0	3	$(1,\ -1/4)$

Table 1. (G_{V_0}, H_{V_0}) for seven supersingular curves at $p = 2$, at the first stage V_0 where $L_p^\flat$ does not vanish (cf. Remark 4.1). Values are recovered from SageMath's `Dp_valued_series` via the bridge formula (15); all seven pairs lie in $\mathbb{Q}^2$ (a consequence of Theorem 3.6 and the rationality of SageMath's modular-symbol output). The sample was chosen to span the full conductor range of the verified set; the remaining 173 curves yield analogous rational (G_{V_0}, H_{V_0})-pairs and are tabulated in the supplementary code repository. A reproducer script is provided at `code/sage_verify_paper_table1.py`.

Remark 4.1 (Definition of V_0 and rationality)**.** We use $V_0 = V_0^\flat$ to denote the first stage $k \geq 0$ at which $L_p^\flat$ is nonzero; equivalently, the smallest index k for which the second component $v_1[k]$ of the SageMath output is nonzero. The Mazur–Tate stage V_0 thus depends on the curve and is in general distinct from $X = 0$ (which is treated in Theorem 3.1); the table entries below are values of L_p at T^{V_0}, not at $T = 0$. The rational form of each entry follows from Theorem 3.6, since SageMath's modular-symbol output is rational and the bridge formula (15) preserves rationality.

Remark 4.2 (Asymmetric vanishing of $L_p^\sharp$ and $L_p^\flat$)**.** In general, $L_p^\sharp$ may vanish to a different order than $L_p^\flat$; let $V_0^\sharp$ and $V_0^\flat = V_0$ denote the respective vanishing levels. The curve `55935g1` provides a concrete instance with $V_0^\sharp = 4$ while $V_0^\flat = 5$: SageMath's $v_0[4] = 1/2 \neq 0$ at one stage before v_1 becomes nonzero. The asymmetric vanishing phenomenon is intrinsic to Sprung's framework and is discussed in [3, Section 6]. For all seven curves in Table 1, the (G_{V_0}, H_{V_0})-pair is recovered at the $L_p^\flat$-vanishing stage $V_0^\flat$.

4.2. **Computational details and code.** All verifications were conducted in SageMath 10 using the full Cremona database (largest conductor 499 998). The bridge formula (15) was implemented as a postprocessing step on the `Dp_valued_series` output, with no Frobenius-precision issues arising in the tested range. The verification code and the complete log files are publicly available at

https://github.com/relkuch-del/sprung-h-matrix-p2.

Class	n	Conductor range	Bit-exact
$a_p = -2$ supersingular	75	[37, 997]	75/75
$a_p = 0$ supersingular (low)	75	[77, 225]	75/75
$a_p = 0$ supersingular (high)	30	[1 001, 63 096]	30/30
Total	**180**	[37, 63 096]	**180/180**

Table 2. Numerical verification of Theorem 3.6: for each curve, the inverse bridge formula (15) yields a rational (G_k, H_k) at every Mazur–Tate stage k, in bit-exact agreement with SageMath's `Dp_valued_series` output. Verified on 180 curves spanning four orders of magnitude in conductor; zero discrepancies.

4.3. **Summary.**

Remark 4.3 (The case $a_p = +2$)**.** The case $a_p = +2$ at $p = 2$ is covered theoretically by Proposition 3.4(iii) and Corollary 3.5(iii), but it does not occur for any of the 180 tested Cremona curves with $\mathrm{rank}(\mathrm{E}/\mathbb{Q}) \geq 1$. We have $\det((1-\varphi)^2) = 1/4$ in this case.

5. Discussion and open questions

5.1. **Relation to Sprung's main theorem at odd primes.** Sprung's main theorem [4] concerns the case of an odd supersingular prime. The case $p = 2$ remains open in the published literature. The explicit identification of Corollary 3.5 and the SageMath bridge of Theorem 3.6 provide concrete computational tools for further study of the $p = 2$ case.

5.2. **Convergence of $\mathcal{H}(X)$ for $X \neq 0$.** The analysis here treats only $X = 0$, where the limit (3) degenerates algebraically by Theorem 3.1. For $X \neq 0$ the cofactor matrices $C_k(X)$ depend nontrivially on k, and explicit power-series expansions of $\mathcal{H}(X)$ in X require the convergence analysis of [3, §4]. We leave the explicit computation of higher-order terms to future work.

5.3. **Higher Mazur–Tate stages.** Theorem 3.6 translates SageMath output to the $(1, \alpha)$-basis at every Mazur–Tate stage k. The resulting sequences $(G_k, H_k)_{k\geq 0}$ display rich rational structure across families of curves, including linearity patterns and stratifications by 2-adic valuation. We do not address these phenomena here; an empirical study of the higher-stage structure is the subject of separate work.

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